

JOSHUA APOLLO

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Computer Engineering student at NTU specialising in systems security and AI infrastructure.

Seeking to apply full-stack and distributed systems experience to security or cloud engineering teams.

EDUCATION

Nanyang Technological University (NTU), Aug 2025 - 2028

B.Comp (Hons) Computer Engineering, **GPA: 4.4/5**

Ngee Ann Polytechnic, Apr 2019 - May 2022

Diploma in Cybersecurity, **GPA: 3.65/4**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Software Engineering, Object-Oriented Programming, Ethical Hacking, Databases, Cryptography, Malware Analysis, Front-End Development

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, Java, JavaScript, Assembly (MIPS/x86)

Frameworks & Tools: Flutter, Restful APIs, Android Studio, FastAPI, ChromaDB, Burp Suite, Metasploit

Key Domains: Retrieval-Augmented Generation (RAG), Offensive Security, Application Security

Platforms: iOS Development, Linux, Azure

PROJECTS

MacroPollo | AI Financial Intelligence Platform | 2026 Creator & Engineer | FastAPI, ChromaDB, Qwen, RAG, Flutter

- Architected an isolated, on-premise AI inference stack using FastAPI and Qwen, delivering real-time macroeconomic risk analysis with zero exposure to third-party APIs, curbing compliance and data leakage risk
- Engineered a Retrieval-Augmented Generation (RAG) pipeline integrated into an iOS app, enabling threat chain mapping against live investment portfolios and reducing manual vulnerability assessment time from hours to minutes
- Designed a zero-data-leakage architecture using ChromaDB as a local vector store, serving as a robust DLP control for proprietary financial data

Rail Energy System | SBS Hackathon | 2026 National Winner, 1st Place, SGD 3,000 | Outcompeted 200+ teams

- Designed a decentralised energy resilience solution for Singapore's national rail network, addressing single points of failure in critical infrastructure
- Conducted end-to-end technical risk assessments, presenting engineering viability directly to C-suite judges

EXPERIENCE

Cybersecurity Intern & Cyber Tools Researcher | Thinkture | Apr 2021 – Feb 2022

- Built custom Python tooling to automate detection and scanning of all OWASP Top 10 vulnerabilities, reducing manual security audit time across client engagements
- Engineered a ransomware proof-of-concept in a controlled lab environment to simulate advanced persistent threat (APT) evasion behaviour, used to stress-test and harden client infrastructure
- Executed mobile and network penetration tests using Burp Suite and Metasploit; translated discovered vulnerabilities into prioritised, business-impact-focused remediation reports for clients

Firefighting Trainer, Sergeant | Singapore Civil Defence Force | Oct 2022 – Aug 2024 | 2-Year Mandatory Service

- Awarded the 2024 Good Performance Award for sustained operational excellence.

Independent Creator & Partnerships Lead | YouTube | Aug 2024 – Mar 2025

- Managed 30+ end-to-end production projects for international partners including Uppbeat and Dehancer.

LEADERSHIP & ACHIEVEMENTS

International CTF Representative | NTU Sentinels | 2025

- Selected to represent NTU at the Vietnam National Cybersecurity Association CTF, competing against university teams internationally: demonstrating applied offensive and defensive security skills under competitive conditions

Subcommittee Member | NTU Sentinels Cybersecurity Club | 2025 – Present

- Co-led planning of SentCTF, NTU's flagship cybersecurity competition; coordinated cross-functional teams and drove industry partnership outreach to expand event reach and sponsorship

Chief Group Leader & Student Ambassador | CCDS Orientation, NTU | 2025 – Present

- Commanded a committee of 10 student leaders across large-scale orientation events, maintaining 100% safety compliance across all operational timelines

President | Debate Society | Jan 2017 – Mar 2018

- Multiple 'Best Speaker' awards for constructing rigorous, policy-driven arguments under competitive pressure